6 **(a)** Fig. 6.1 is a full-size diagram of a lens and an image I of an object.

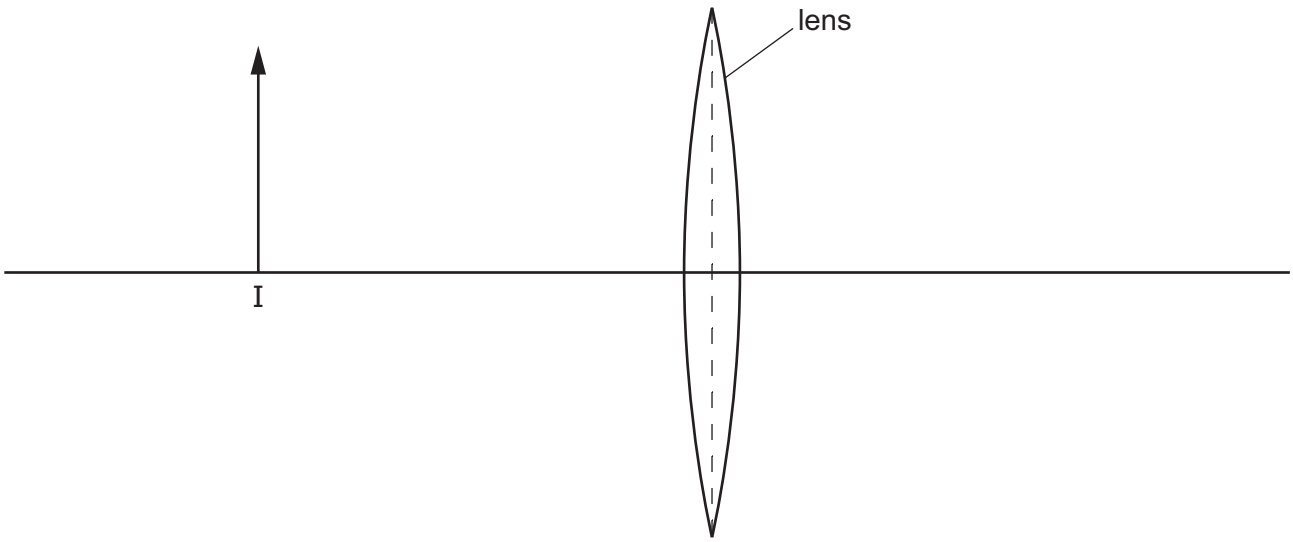

Fig. 6.1

The focal length of the lens is 3.0 cm.

(i) On Fig. 6.1, mark **two** points with labels F, to show the positions of the principal focuses (focal points) of the lens. [1]

(ii) On Fig. 6.1, draw **two** rays from the image I to locate the object. Draw the object and label it O. [3]

(iii) The object is moved so that it is 6 cm to the left of the lens.

State **one** characteristic of the image formed when the object is in this position.

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Fig. 6.2 shows a simplified diagram of an eye with rays from a near object. The eye needs correction for long-sightedness.

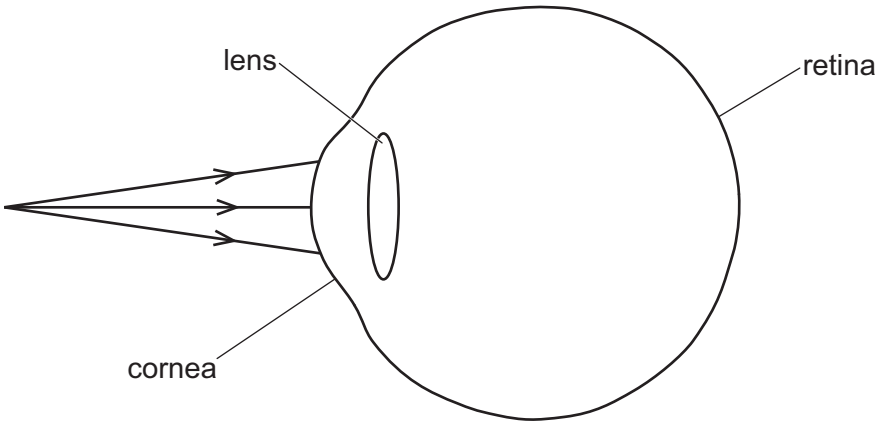

Fig. 6.2

(i) On Fig. 6.2, draw the path of the rays inside the eye to show the effect of long-sightedness. [1]

(ii) A lens is used to correct the long-sightedness.

Draw a lens suitable for this correction.

lens to correct long-sightedness

[1]

[Total: 7]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN